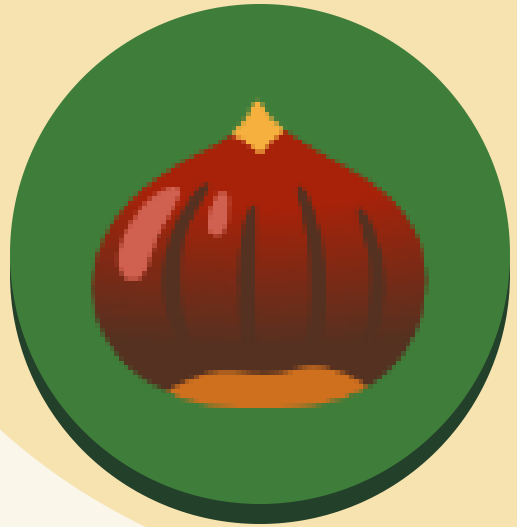
Horticulture 1 · Unit 6 · Level 2 (modified)

Seeds



TODAY'S PLAN

- 🌰 Parts of a seed
- 🌱 Germination
- 🔨 Special treatments
- 🪴 Direct vs. indirect seeding

Review · 5 minutes

Quick Review



🟢 What do N, P and K do?

🫙 Which soil layer settles first?

🪴 Why use potting mix in the greenhouse?

Words to Know

Words to Know



Seed

A plant embryo with stored food inside a protective seed coat.



Seed coat

The hard outer covering that protects the seed.



Embryo

The tiny plant inside the seed that grows into the new plant.



Endosperm

Stored food inside the seed that feeds the embryo as it sprouts.

Words to Know

More Words to Know



🌱 **Germination**: When a seed takes in water and begins to grow a root and shoot.

🍃 **Cotyledon**: Seed leaves that come up first and feed the seedling. Monocots have 1, dicots have 2.

🌿 **True leaves**: Leaves that grow after the cotyledons and look like the adult plant's leaves.

🪴 **Direct seeding**: Sowing seeds right where the plant will grow.

Learn

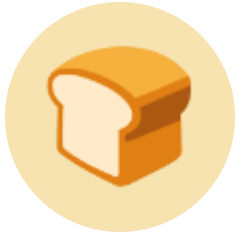
Parts of a Seed



Seed coat
protects



Embryo
becomes the plant

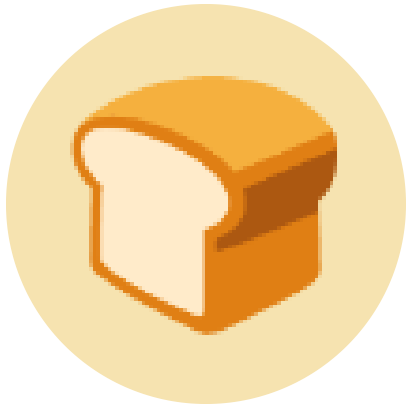


Endosperm
stored food



Cotyledons
seed leaves

🃏 Check 1 of 3



Which part feeds the sprouting embryo?

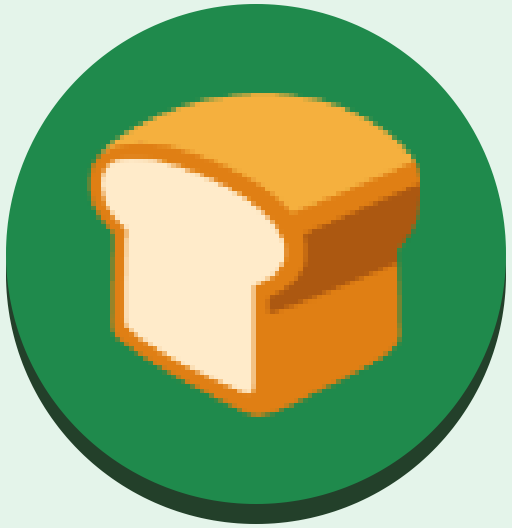
Hold up 1, 2, or 3:

1️⃣ Seed coat

2️⃣ Endosperm

3️⃣ True leaves

✅ Answer · Card 1 of 3



Endosperm

Stored food.

Learn

Germination



Seeds need **water, warmth and air** (some need light, some need dark).

First the **root (radicle)** grows down, then the shoot grows up.

Cotyledons open first, then **true leaves**.

Compare

Monocot vs. Dicot



Monocot

1 cotyledon
corn, grasses, lilies

Dicot

2 cotyledons
beans, sunflowers

Learn

Special Treatment of Seeds



Some seeds have very hard coats.

Scarification: nick or sand the coat so water gets in.

Soaking and **cold stratification** (chilling) help others sprout.

🃏 Check 2 of 3



Scarification means…

Hold up 1, 2, or 3:

1️⃣ Scaring birds

2️⃣ Planting deep

3️⃣ Scratching a hard seed coat so water enters

✅ Answer · Card 2 of 3



Scratching a hard seed coat so water enters

Helps germination.

Compare

Seeding Methods



Direct seeding

Sow where it will grow

beans, carrots, corn

Indirect seeding

Start in trays, transplant later

tomatoes, petunias

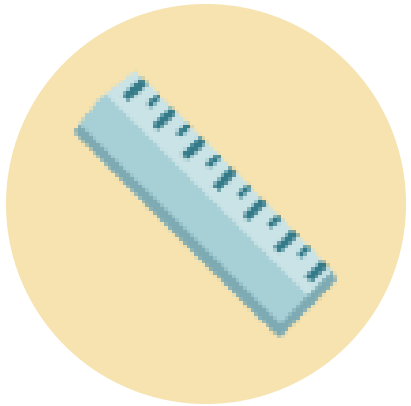
Steps

Sowing in Trays



1. Fill tray with moist seed-starting mix
2. Plant 2-3 times as deep as the seed is wide
3. Label with name and date
4. Water gently; keep warm and moist

🃏 Check 3 of 3



How deep do you plant a seed?

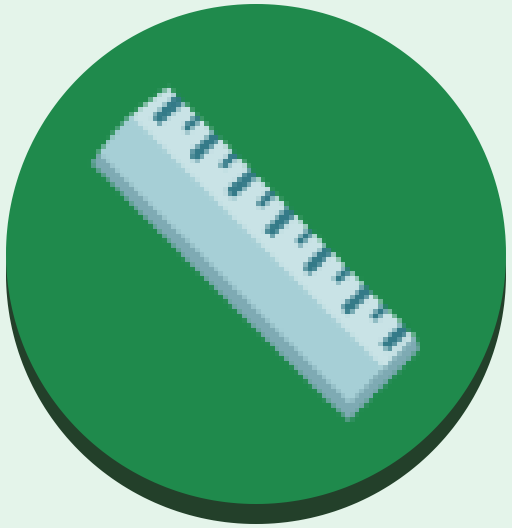
Hold up 1, 2, or 3:

1️⃣ 2–3 times its width

2️⃣ As deep as possible

3️⃣ On top of a rock

✅ Answer · Card 3 of 3



2–3 times its width

Small seeds go shallow.

Wrap-Up

Recap



Seed parts
coat, embryo, endosperm



Germination
water, warmth, air



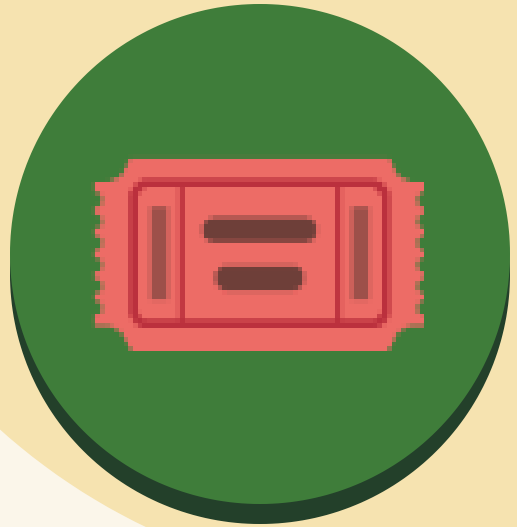
Scarify
nick hard coats



Indirect
trays, transplant

Wrap-Up

Exit Ticket



Name the 3 main parts of a seed.

What 3 things do seeds need to germinate?

Direct vs. indirect seeding: give one example each.